Topic 6: Frequency Response of FIR Filters

Summary based on J. H. McClellan, R. W. Schafer, and M. A. Yoder, Signal Processing First, Prentice Hall, 2003.

Topics: sinusoidal response of FIR systems, superposition, transient and steady-state behaviour, frequency-response properties, magnitude and phase, delay and first-difference filters, cascaded systems, moving-average filters, and filtering sampled continuous-time signals.

> **Key idea.** For an LTI FIR filter, each sinusoidal input component keeps its frequency while its amplitude and phase are modified by the filter's frequency response. This makes the frequency response a compact description of how the filter treats every spectral component.

6.1 Sinusoidal response of FIR systems

Consider the causal FIR filter

$$y[n] = \sum_{k=0}^{M} b_k\, x[n-k]$$

and the complex-exponential input

$$x[n] = Ae^{j(\hat{\omega}n+\phi)}.$$

Substitution into the difference equation gives

$$\begin{aligned} y[n] \quad &= \sum_{k=0}^{M} b_k\, Ae^{j[\hat{\omega}(n-k)+\phi]} \\ &= Ae^{j(\hat{\omega}n+\phi)} \sum_{k=0}^{M} b_k\, e^{-j\hat{\omega}k}. \end{aligned}$$

Define

$$H\left(e^{j\hat{\omega}}\right) = \sum_{k=0}^{M} b_k\, e^{-j\hat{\omega}k} = \sum_{k=-\infty}^{\infty} h\,[k]e^{-j\hat{\omega}k}.$$

The output is therefore

$$y[n] = H\left(e^{j\hat{\omega}}\right)x[n].$$

Writing the frequency response in polar form,

$$H\left(e^{j\hat{\omega}}\right) = \left|H\left(e^{j\hat{\omega}}\right)\right| e^{j\angle H\left(e^{j\hat{\omega}}\right)},$$

we obtain

$$y[n] = A\left|H\left(e^{j\hat{\omega}}\right)\right| e^{j\left[\hat{\omega}n+\phi+\angle H\left(e^{j\hat{\omega}}\right)\right]}.$$

Thus, the filter multiplies the input amplitude by $\left|H\left(e^{j\hat{\omega}}\right)\right|$ and adds $\angle H\left(e^{j\hat{\omega}}\right)$ to the input phase.

Example 6.1. For the three-point moving average,

$$y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2]),$$

$$H(e^{j\hat{\omega}}) = \frac{1}{3}\left(1 + e^{-j\hat{\omega}} + e^{-j2\hat{\omega}}\right).$$

At $\hat{\omega} = 0$, $H = 1$, so DC passes unchanged. At $\hat{\omega} = 2\pi/3$, the three phasors sum to zero, so that sinusoidal component is completely removed.

6.1 Superposition and frequency response

If the input is a sum of complex exponentials,

$$x[n] = \sum_{r=1}^{R} A_r\, e^{j(\hat{\omega}_r n + \phi_r)},$$

then linearity gives

$$y[n] = \sum_{r=1}^{R} A_r\, H(e^{j\hat{\omega}_r}) e^{j(\hat{\omega}_r n + \phi_r)}.$$

Each spectral component is treated independently: its frequency is unchanged, while its amplitude and phase are modified according to the frequency response at that frequency.

Example 6.1. Suppose

$$x[n] = 2\cos(0.2\pi n) + \cos(0.8\pi n)$$

is applied to a low-pass FIR filter for which

$$|H(e^{j0.2\pi})| = 0.95, \qquad |H(e^{j0.8\pi})| = 0.10.$$

Ignoring phase for simplicity, the output amplitudes are approximately 1.90 and 0.10. The low-frequency component is preserved, while the high-frequency component is strongly attenuated.

6.3 Steady-state and transient response

The sinusoidal-response result above assumes a complex exponential extending for all n. In practice, a sinusoid may begin at $n = 0$:

$$x[n] = e^{j\hat{\omega}n} u[n].$$

For an FIR filter,

$$y[n] = \sum_{k=0}^{M} b_k\, e^{j\hat{\omega}(n-k)} u[n-k].$$

Since $u[n-k] = 0$ when $k > n$,

$$y[n] = e^{j\hat{\omega}n} \sum_{k=0}^{\min(M,n)} b_k\, e^{-j\hat{\omega}k}.$$

Three regions can be distinguished:

1. **Before the input begins:** the output is zero.
2. **Transient region:** during the first M samples, the averaging window is only partly filled.
3. **Steady-state region:** for $n \geq M$, the output is

$$y[n] = H(e^{j\hat{\omega}}) e^{j\hat{\omega}n}.$$

When a finite-duration input ends, another transient of approximately M samples occurs.

6.2 Properties of the frequency response

6.2.1 Relation to impulse response and difference equation

The same FIR filter can be described in the time and frequency domains:

$$y[n] = \sum_{k=0}^{M} h[k]x[n-k]$$

corresponds to

$$Y(e^{j\hat{\omega}}) = H(e^{j\hat{\omega}})X(e^{j\hat{\omega}}),$$

where

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^{M} h[k]e^{-j\hat{\omega}k}.$$

6.2.2 Periodicity

Discrete-time frequency response is periodic with period 2π:

$$H(e^{j(\hat{\omega}+2\pi)}) = H(e^{j\hat{\omega}}).$$

It is therefore sufficient to specify one interval, commonly

$$-\pi \leq \hat{\omega} < \pi.$$

6.2.3 Conjugate symmetry

For real-valued impulse responses,

$$H(e^{-j\hat{\omega}}) = H^*(e^{j\hat{\omega}}).$$

Consequently,

$$|H(e^{-j\hat{\omega}})| = |H(e^{j\hat{\omega}})|$$

and

$$\angle H(e^{-j\hat{\omega}}) = -\angle H(e^{j\hat{\omega}}).$$

The magnitude is even and the phase is odd, apart from possible 2π phase jumps.

6.5 Graphical representation of frequency response

A frequency-response plot normally has two parts:

- the **magnitude response**, $|H(e^{j\hat{\omega}})|$, showing amplification or attenuation;
- the **phase response**, $\angle H(e^{j\hat{\omega}})$, showing phase shift or delay.

6.5.1 Delay system

A delay of n_0 samples is

$$y[n] = x[n-n_0],$$

with impulse response

$$h[n] = \delta[n - n_0].$$

Its frequency response is

$$H(e^{j\widehat{\omega}}) = e^{-j\widehat{\omega}n_0}.$$

Therefore,

$$|H(e^{j\widehat{\omega}})| = 1, \qquad \angle H(e^{j\widehat{\omega}}) = -\widehat{\omega}n_0.$$

The system changes no amplitude; it introduces only a linear phase shift.

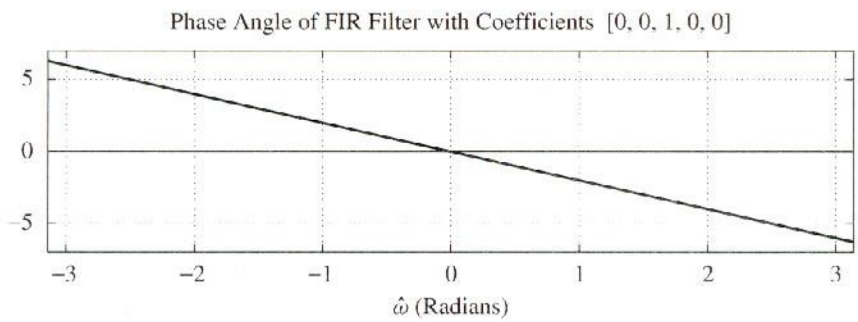

Figure 6.1. Phase response of a delay system with n_0 = 2 .

6.5.2 First-difference system

Consider

$$y[n] = x[n] - x[n - 1].$$

Then

$$H(e^{j\widehat{\omega}}) = 1 - e^{-j\widehat{\omega}} = 2je^{-j\widehat{\omega}/2}\sin\left(\frac{\widehat{\omega}}{2}\right).$$

Hence,

$$|H(e^{j\widehat{\omega}})| = 2\left|\sin\left(\frac{\widehat{\omega}}{2}\right)\right|.$$

The response is zero at $\widehat{\omega} = 0$, so constant signals are removed. Higher frequencies receive greater gain; the system therefore behaves as a simple high-pass filter and as a discrete differentiator.

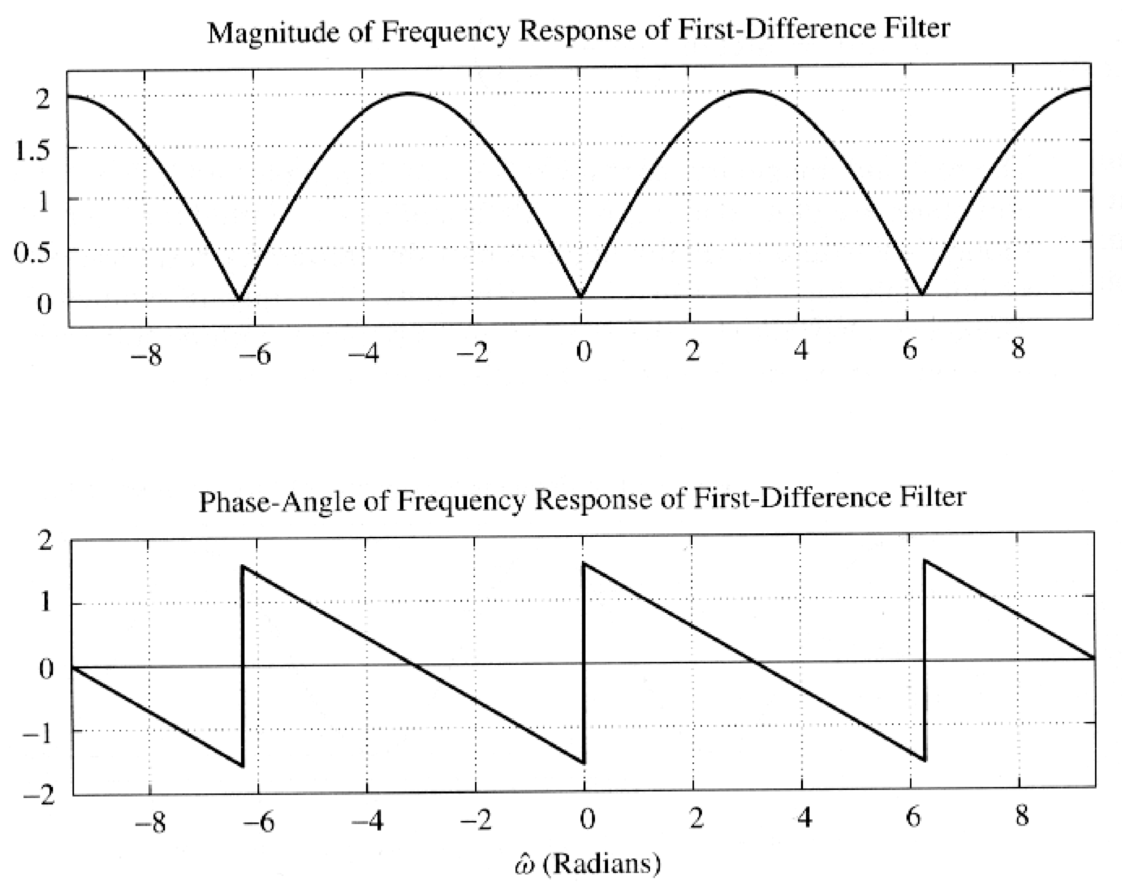

Figure 6.2. Magnitude and phase of the first-difference filter over three periods.

Example 6.3. For the unit-step input $x[n] = u[n]$,

$$y[n] = u[n] - u[n-1] = \delta[n].$$

The first-difference filter responds only at the transition. It is therefore useful for detecting abrupt changes and edges.

6.6 Cascaded LTI systems

For two LTI systems in cascade,

$$H_{\text{eq}}(e^{j\hat{\omega}}) = H_1(e^{j\hat{\omega}})H_2(e^{j\hat{\omega}}).$$

Because multiplication is commutative, the cascade order does not change the overall input-output response:

$$H_1H_2 = H_2H_1.$$

In the time domain, the corresponding relation is

$$h_{\text{eq}}[n] = h_1[n] * h_2[n].$$

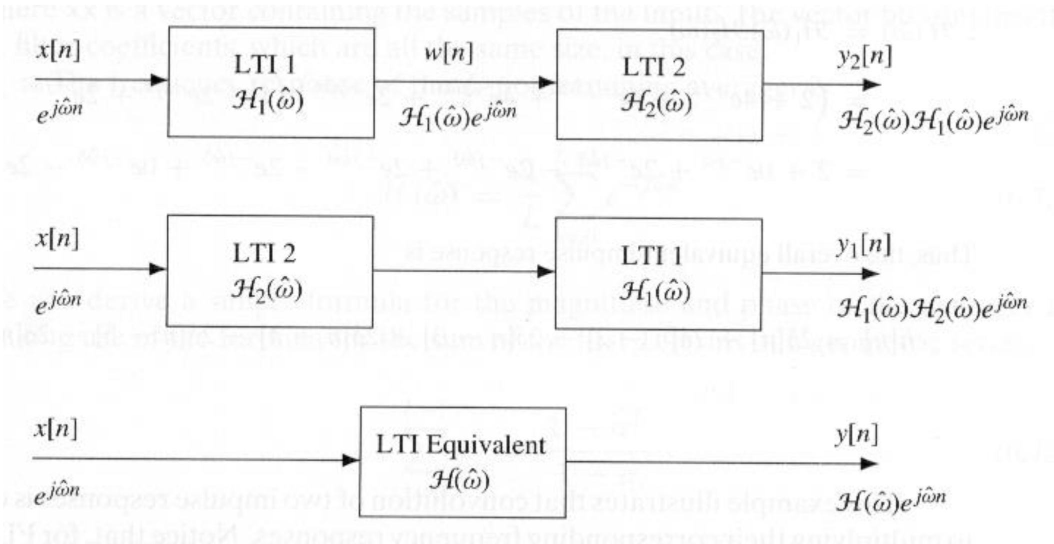

Figure 6.3. Three equivalent representations of cascaded LTI systems.

This gives the useful correspondence

convolution in time $\Leftrightarrow$ multiplication in frequency.

Example 6.2. Cascading two identical two-point moving averages gives

$$H_1\left(e^{j\hat{\omega}}\right) = H_2\left(e^{j\hat{\omega}}\right) = \frac{1}{2}\left(1 + e^{-j\hat{\omega}}\right),$$

so

$$H_{\text{eq}}\left(e^{j\hat{\omega}}\right) = \frac{1}{4}\left(1 + 2e^{-j\hat{\omega}} + e^{-j2\hat{\omega}}\right).$$

The equivalent impulse response is

$$h_{\text{eq}}[n] = \left\{\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right\},$$

which is a three-point triangular smoother.

6.3 Moving-average filtering

An L-point causal moving-average filter is

$$y[n] = \frac{1}{L}\sum_{k=0}^{L-1} x\,[n-k].$$

Its impulse response is

$$h[k] = \begin{cases} \frac{1}{L}, & 0 \le k \le L-1, \\ 0, & \text{otherwise.} \end{cases}$$

The frequency response is

$$H\left(e^{j\hat{\omega}}\right) = \frac{1}{L}\sum_{k=0}^{L-1} e^{-j\hat{\omega}k}.$$

Using the finite geometric-series formula,

$$\sum_{k=0}^{L-1} r^k = \frac{1-r^L}{1-r},$$

with $r = e^{-j\hat{\omega}}$,

$$H(e^{j\hat{\omega}}) = \frac{1}{L}\frac{1-e^{-jL\hat{\omega}}}{1-e^{-j\hat{\omega}}} = e^{-j\hat{\omega}(L-1)/2}\frac{\sin(L\hat{\omega}/2)}{L\sin(\hat{\omega}/2)}.$$

Thus,

$$|H(e^{j\hat{\omega}})| = \left|\frac{\sin(L\hat{\omega}/2)}{L\sin(\hat{\omega}/2)}\right|$$

and, between phase discontinuities,

$$\angle H(e^{j\hat{\omega}}) = -\frac{L-1}{2}\hat{\omega}.$$

The response has zeros at

$$\hat{\omega} = \frac{2\pi m}{L}, \qquad m = 1,2,\dots,L-1.$$

A longer average produces a narrower main lobe and stronger smoothing, but also a larger delay of $(L-1)/2$ samples.

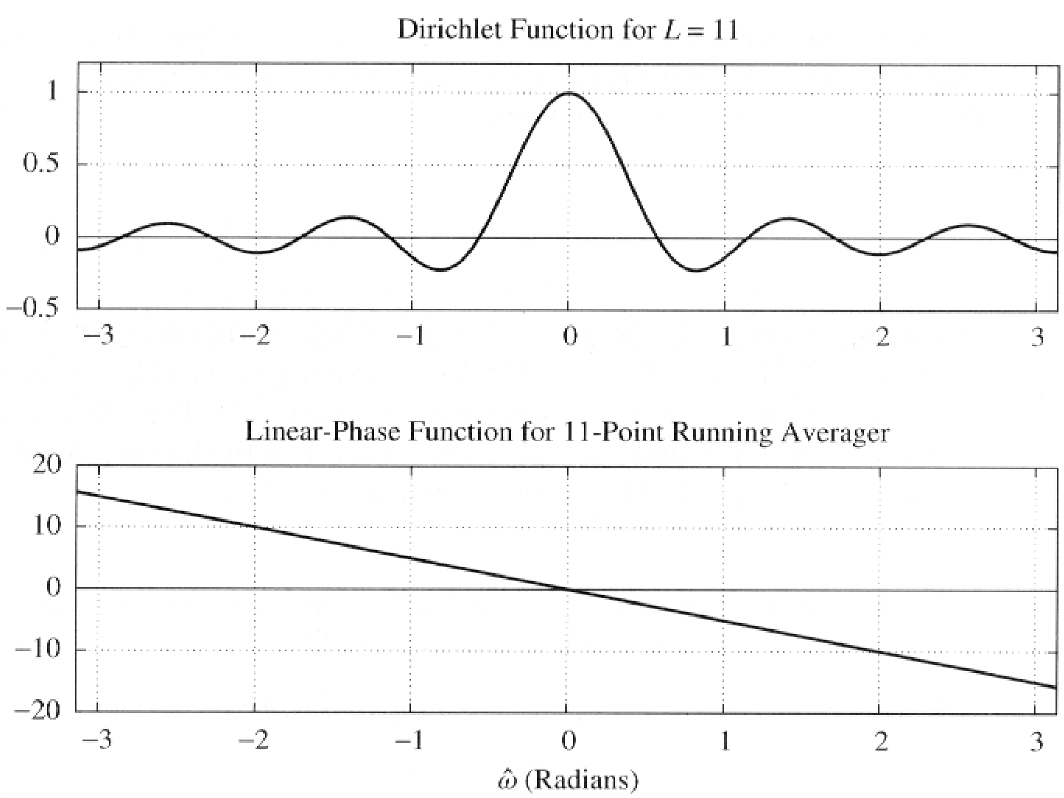

Figure 6.4. Amplitude factor and linear-phase factor for an 11-point moving-average filter.

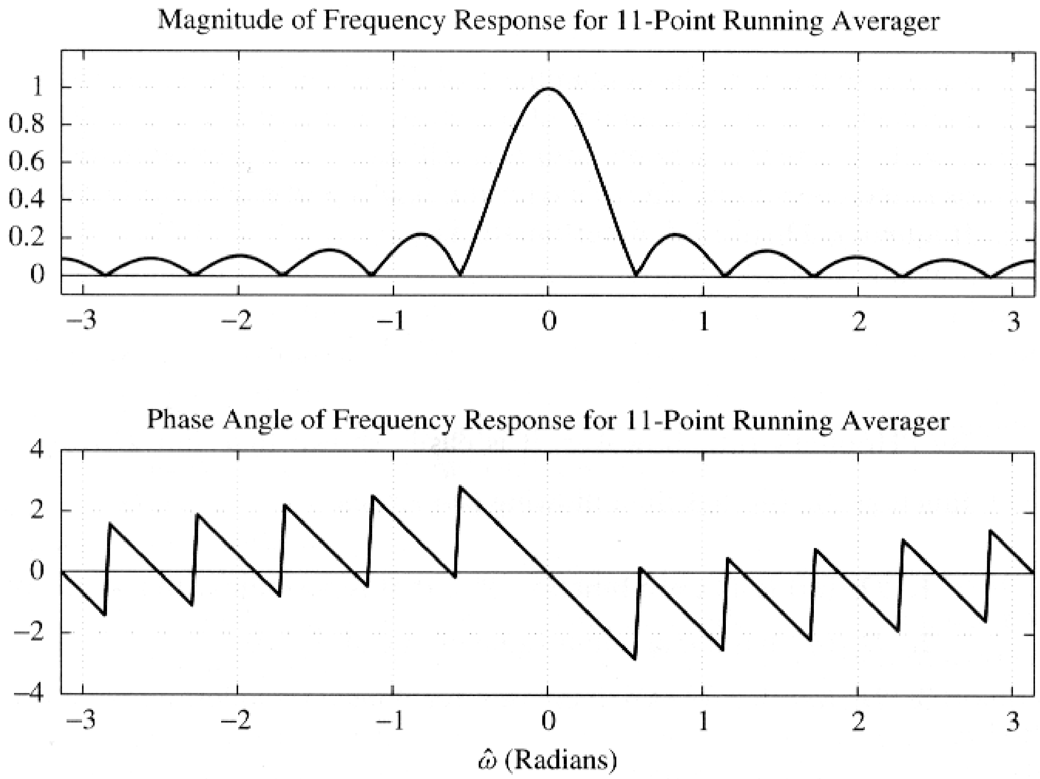

Figure 6.5. Magnitude and phase response of an 11-point moving-average filter.

Example 6.5. For $L = 11$, the first positive zero occurs at

$$\hat{\omega} = \frac{2\pi}{11}.$$

A sinusoid at this normalized frequency is cancelled exactly. Low-frequency components near DC pass with gain close to one, while many higher-frequency components are attenuated. The filter is therefore a simple low-pass smoother.

6.3.2 Worked example: filtering a multicomponent signal

Consider the three-point symmetric FIR filter with coefficients {1, 2, 1}. Its frequency response is

$$\mathcal{H}(\hat{\omega}) = 1 + 2e^{-j\hat{\omega}} + e^{-j2\hat{\omega}} = (2 + 2\cos\hat{\omega})e^{-j\hat{\omega}}$$

For the input

$$x[n] = 4 + 3\cos\left(\frac{\pi}{3}n - \frac{\pi}{2}\right) + 3\cos\left(\frac{20\pi}{21}n\right)$$

the response is evaluated separately at each input frequency. The required values are

$$\mathcal{H}(0) = 4, \qquad \mathcal{H}\left(\frac{\pi}{3}\right) = 3e^{-j\pi/3}, \qquad \mathcal{H}\left(\frac{20\pi}{21}\right) = 0.0223e^{-j20\pi/21}$$

Therefore,

$$y[n] = 16 + 9\cos\left(\frac{\pi}{3}(n-1) - \frac{\pi}{2}\right) + 0.067\cos\left(\frac{20\pi}{21}(n-1)\right)$$

The DC term is multiplied by four, the component at π/3 is amplified by three and delayed by one sample, and the component close to π is almost completely suppressed. This example shows how the magnitude-response plot can be read directly as a frequency-dependent gain.

6.3.3 Worked example: transient and steady-state response

Take the FIR filter with coefficients {1, −2, 4, −2, 1} and the suddenly applied input

$$x[n] = cos(0.2\pi n - \pi)u[n]$$

The filter order is M=4, so the initial transient lasts four samples. After that interval, the output has reached its steady-state sinusoidal form. Evaluating the frequency response at $\hat{\omega}$=0.2π gives

$$H(0.2\pi) = 1.382e\hat{}(-j0.4\pi)$$

and hence, for n≥4,

$$y\_ss[n] = 1.382cos(0.2\pi(n-2) - \pi)$$

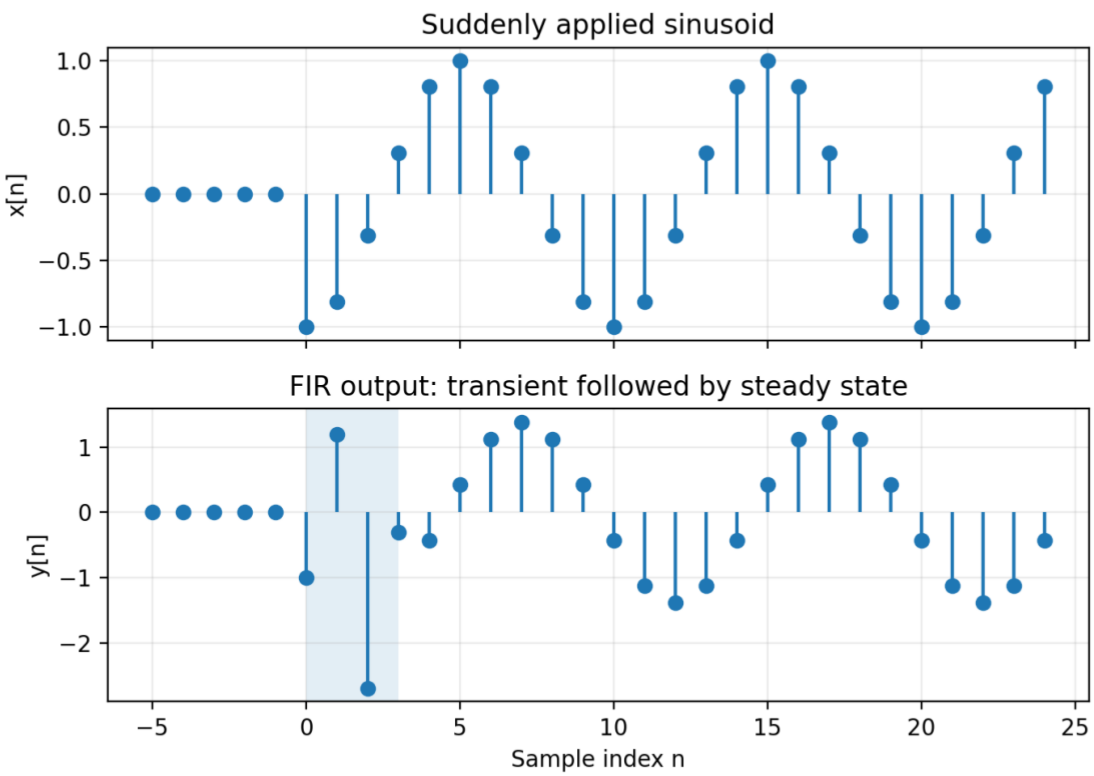

Figure 6.6. Suddenly applied sinusoid: initial transient and steady-state FIR response.

The shaded interval in the output plot is the transient region. The steady-state output is a scaled version of the input delayed by two samples, but this simple sinusoidal description is not valid during the first four samples.

6.4 Filtering sampled continuous-time signals

When an analogue sinusoid is sampled, processed by a discrete-time LTI filter, and ideally reconstructed, its analogue frequency is preserved provided that the sampling theorem is satisfied. For an input complex exponential x(t)=Xe^(jωt), the sampled normalized frequency is

$$\hat{\omega} = \omega T\_s = 2\pi f/f\_s$$

and the reconstructed output is

$$y(t) = H(\omega T\_s) X e\^(j\omega t)$$

Thus, over the nonaliased analogue-frequency interval, the equivalent continuous-time frequency response is H(ωT_s).

6.4.1 Example: an 11-point averager at 1 kHz

Let f_s=1000 samples/s and use an 11-point moving-average filter. Consider

$$x(t) = cos(2\pi \cdot 25t) + sin(2\pi \cdot 250t)$$

The two normalized frequencies are 0.05π and 0.5π. Evaluating the frequency response gives

$$H(0.05\pi) = 0.8811 e\^(-j\pi/4), \quad H(0.5\pi) = 0.0909 e\^(-j\pi/2)$$

The reconstructed output is therefore

$$y(t) = 0.8811 cos(2\pi \cdot 25t - \pi/4) + 0.0909 sin(2\pi \cdot 250t - \pi/2)$$

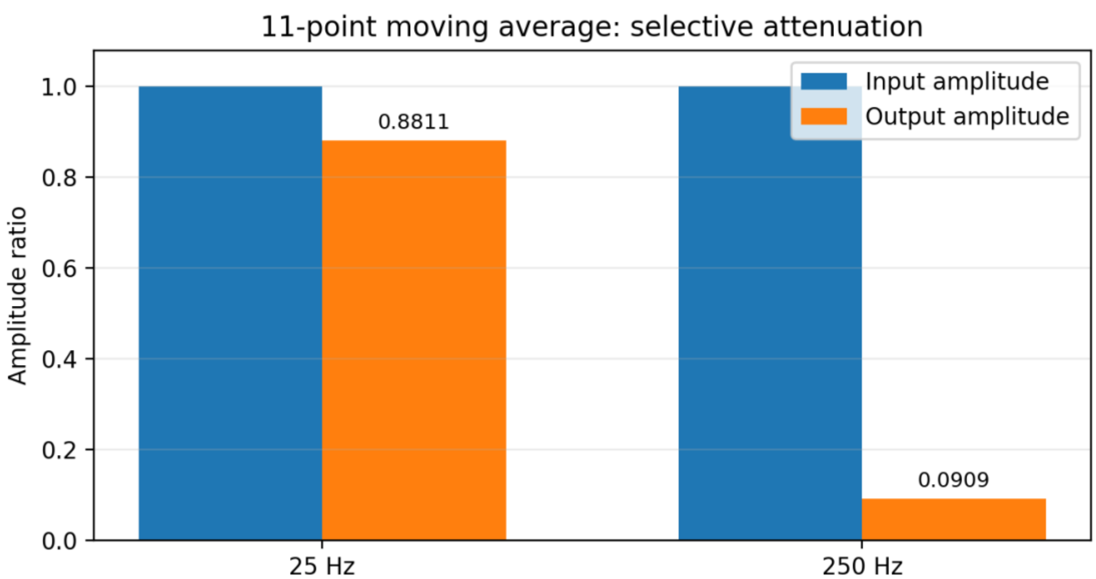

Figure 6.7. The 11-point moving average preserves the 25-Hz component much more strongly than the 250-Hz component.

The 250-Hz component is strongly attenuated because it lies in a low-gain part of the moving-average response. This is the continuous-time interpretation of the filter's discrete-time low-pass behaviour.

6.4.2 Interpreting the delay of a moving average

The phase factor of an L-point moving average is e^(−jω̂(L−1)/2), corresponding to a delay of (L−1)/2 samples. For odd L this is an integer delay; for even L it is a half-integer delay.

$$\tau\_d = ((L-1)/2) T\_s$$

At ω̂=0.2π, a five-point average has gain 0.6472 and a delay of two samples, while a four-point average has gain 0.7694 and a delay of 1.5 samples.

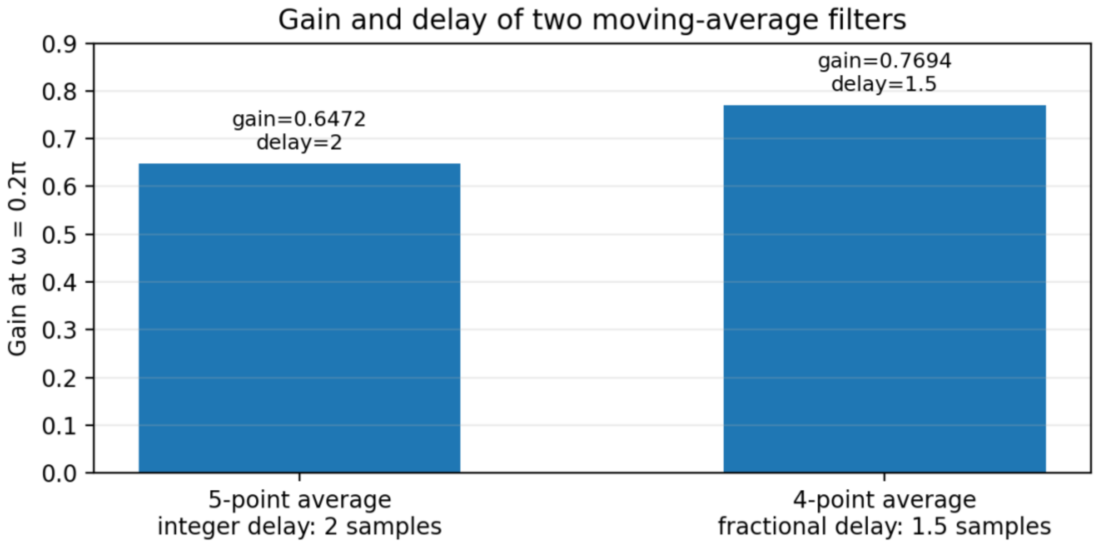

Figure 6.8. Comparison of gain and delay for five-point and four-point moving averages.

A fractional-sample delay cannot be seen merely as shifting the discrete samples by an integer. After ideal reconstruction, however, it corresponds to an ordinary continuous-time delay of 1.5T_s.

6.9Key points

1. Complex exponentials are eigenfunctions of LTI systems: the output frequency is unchanged.
2. $H\left(e^{j\hat{\omega}}\right)$ is the discrete-time Fourier transform of $h[n]$.
3. Magnitude controls amplitude; phase controls phase shift and delay.
4. For real FIR filters, magnitude is even and phase is odd.
5. Cascading systems multiplies their frequency responses.
6. Moving averages are low-pass filters whose smoothing and delay increase with filter length.

6.10 Notation

Symbol	Meaning
x[n], y[n]	Input and output discrete-time sequences
h[n]	Impulse response of the FIR filter
b_k	FIR filter coefficient at delay k
M	Filter order; the largest delay index
L=M+1	Filter length or number of coefficients
H(e^{jω̂})	Frequency response evaluated at normalized radian frequency ω̂
|H(e^{jω̂})|	Magnitude response or frequency-dependent gain
∠H(e^{jω̂})	Phase response
ω̂	Normalized radian frequency, in radians per sample

ω	Continuous-time radian frequency, in radians per second
T_s	Sampling period
$f_s=1/T_s$	Sampling rate
*	Convolution operator
$\delta[n]$	Unit impulse sequence
$u[n]$	Unit-step sequence